# SURJECTIVITY OF A MORPHISM FROM A SCHEME EQUALS SURJECTIVITY ON TOPOLOGICAL SPACES

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

*Source reference.* Original reference 025V: sources/stacks-chow/tags.txt, source lines 2828–2828. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $f : T \to X$ be a morphism from a scheme to $X$. The following are equivalent*

(1) *$f : T \to X$ is surjective (according to Spaces, Definition 025V), and*
(2) *$|f| : |T| \to |X|$ is surjective.*

*Proof.* Assume (1). Let $x : \mathrm{Spec}(K) \to X$ be a morphism from the spectrum of a field into $X$. By assumption the morphism of schemes $\mathrm{Spec}(K) \times_X T \to \mathrm{Spec}(K)$ is surjective. Hence there exists a field extension $K'/K$ and a morphism $\mathrm{Spec}(K') \to \mathrm{Spec}(K) \times_X T$ such that the left square in the diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) \times_X T & \longrightarrow & T \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(K) & = & \mathrm{Spec}(K) & \xrightarrow{x} & X \end{array}$$

is commutative. This shows that $|f| : |T| \to |X|$ is surjective.

Assume (2). Let $Z \to X$ be a morphism where $Z$ is a scheme. We have to show that the morphism of schemes $Z \times_X T \to T$ is surjective, i.e., that $|Z \times_X T| \to |Z|$ is surjective. This follows from (2) and Lemma 03H4. □

*Date*: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.